



**SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE
(AUTONOMOUS)**

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

Accredited by NAAC with 'A+' Grade.

Recognised as Scientific and Industrial Research Organisation

SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P.,
INDIA

Regulation R23		IV/IV – B.Tech I - Semester							
INFORMATION TECHNOLOGY									
COURSE STRUCTURE (With effect from 2023-24 affiliated batch onwards)									
Course Code	Course Name	Category	L	T	P	Cr	C.I.E	S.E.E	Total Marks
B23IT4101	Internet of Things	PC	3	0	0	3	30	70	100
B23HS4101	Human Resource & Project Management	HS	2	0	0	2	30	70	100
# PE-IV	Professional Elective - IV	PE	3	0	0	3	30	70	100
# PE-V	Professional Elective - V	PE	3	0	0	3	30	70	100
# OE-III	Open Elective – III	OE	3	0	0	3	30	70	100
# OE-IV	Open Elective – IV	OE	3	0	0	3	30	70	100
B23IT4110	Prompt Engineering	SEC	0	1	2	2	30	70	100
B23MC4101	Constitution of India	MC	2	0	0	-	30	-	30
B23IT4111	Evaluation of Industry Internship /Mini Project	PR	-	-	-	2	--	50	50
Total			19	1	2	21	240	540	780

	Course Code	Course
#PE- IV	B23CS4102	Software Architecture & Design Patterns
	B23IT4102	Deep Learning
	B23IT4103	Computer Vision
	B23IT4104	Blockchain Technology & Applications
	B23IT4105	12 week MOOC Swayam/NPTEL course recommended by the BoS
#PE- V	B23CS4107	Agile Methodologies
	B23IT4106	Big Data Analytics
	B23IT4107	Mobile Computing
	B23IT4108	Cyber Physical Systems
	B23IT4109	12 week MOOC Swayam/NPTEL course recommended by the BoS
#OE-III	Student has to study one Open Elective offered by CE or ECE or EEE or ME or S&H from the list enclosed.	
#OE-IV	Student has to study one Open Elective offered by CE or ECE or EEE or ME or S&H from the list enclosed.	

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT4101	PC	3	--	--	3	30	70	3 Hrs.
INTERNET OF THINGS								
(For IT)								
Course Objectives: This course is designed to:								
1.	Provide a clear understanding of IoT concepts, architecture, enabling technologies, and networking components.							
2.	Explain the working of IoT systems using sensors, actuators, and real-time data processing.							
3.	Introduce IoT communication technologies such as Zigbee, Bluetooth, Wi-Fi, and LPWAN.							
4.	Provide knowledge of IoT frameworks, interoperability, fog computing, and modern system approaches.							
5.	Enable development and understanding of IoT applications in areas like agriculture, vehicles, and smart cities.							
Course Outcomes: Upon the completion of the course students will be able to:								
S.No	Outcome						Knowledge Level	
1.	Understanding of IoT concepts, architecture, enabling technologies, and networking components.						K2	
2.	Demonstrate the working of IoT sensing, actuation, and processing mechanisms in real-time environments.						K3	
3.	Apply IoT connectivity and communication technologies such as Zigbee, Bluetooth, Wi-Fi, and LPWAN in system design.						K3	
4.	Apply IoT interoperability, fog computing frameworks, and emerging paradigms for efficient data processing.						K3	
5.	Demonstrate real-world IoT applications in domains such as agriculture, vehicular systems, and smart cities.						K3	
SYLLABUS								
UNIT-I	Predecessors of IoT: Introduction, Wireless Sensor Networks, Machine-to- Machine Communications, Cyber Physical Systems Emergence of IoT: Introduction, Evolution of IoT, Enabling IoT and the Complex Interdependence of Technologies, IoT Networking Components, Addressing Strategies in IoT.							
UNIT-II	IoT Sensing and Actuation: Introduction, Sensors, Sensor Characteristics, Sensorial Deviations, Sensing Types, Sensing Considerations, Actuators, Actuator Types, Actuator Characteristics. IoT Processing Topologies and Types: Data Format, Importance of Processing in IoT, Processing Topologies, IoT Device Design and Selection Considerations, Processing Offloading.							

UNIT- III	IoT Connectivity Technologies: Introduction, IEEE 802.15.4, Zigbee, Thread, ISA100.11A, WirelessHART, RFID, NFC,DASH7, Z-Wave, Weightless, Sigfox, LoRa, NB-IT, Wi-Fi, Bluetooth IoT Communication Technologies: Introduction, Infrastructure Protocols, Discovery Protocols, Data Protocols, Identification Protocols, Device Management, Semantic Protocols.
UNIT- IV	IoT Interoperability: Introduction, Standards, Frameworks Fog Computing and Its Applications: Introduction, View of Fog Computing Architecture, Fog Computing in IoT, Selected Applications of Fog Computing.
UNIT- V	Paradigms, Challenges, and the Future: Introduction, Evolution of New IoT Paradigms, Challenges Associated with IoT, Emerging Pillars of IoT IoT Case Studies: Agricultural IoT, Vehicular IoT.
Textbooks:	
1	Introduction to IoT, Sudip Misra, Anandarup Mukhaerjee, Arjit Roy, Cambridge University Press, 2021.
2	Internet of Things: Architecture, Design Principles and Applications, Rajkamal, McGraw Hill Higher Education
Reference Books:	
1	Fog and Edge Computing: Principles and Paradigms, Rajkumar Buyya (Editor), Satish Narayana Srirama (Editor) , ISBN: 978-1-119-52498-4, January 2019
2	Getting Started with the Internet of Things, CunoPfister , Oreilly

Estd. 1980

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Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23HS4101	HS	2	--	--	2	30	70	3 Hrs.
HUMAN RESOURCE & PROJECT MANAGEMENT								
(Common to CSE, AIML, IT, AIDS, CSG, CSIT)								
Course Objectives:								
1.	Explain the nature, scope, functions of HRM, and processes of HR planning, recruitment, and selection.							
2.	Explain HRD concepts, training methods, performance appraisal systems, and career development practices.							
3.	Explain the basic concepts of project management, including project life cycle, resource management, and monitoring techniques.							
4.	Explain project selection and appraisal methods using financial and strategic evaluation tools.							
5.	Explain project implementation processes, organizational structures, and performance evaluation methods.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Explain HRM functions, HR planning, and recruitment–selection procedures.							K2
2.	Explain HRD practices, training techniques, and performance appraisal methods.							K2
3.	Explain project management concepts, project networks, and monitoring processes.							K2
4.	Explain project appraisal techniques such as SWOT analysis, payback period, and NPV.							K2
5.	Explain project implementation strategies, organizational forms, and performance evaluation processes.							K2
SYLLABUS								
UNIT-I (10Hrs)	HRM: Nature, Scope, Concept of HRM - Functions of HRM - Role of HR manager – Emerging trends in HRM – E-HRM - HR audit models – ethical aspects of HRM. HR Planning – Job Design- Recruitment - Sources of recruitment -Selection- Selection Procedure.							
UNIT-II (10 Hrs)	HRD: HR accounting – Models - Concept of Training and Development – Methods of Training. Performance Appraisal: Importance – Methods of performance appraisal – Career Development and Counseling – group interaction.							
UNIT-III (10 Hrs)	Basics of Project Management: Concept–resource management- Project environment – Types of Projects –project networks-DPR- Project life cycle – Project proposals –							

	Monitoring project progress – Project appraisal and Project selection-80-20 rules.
UNIT-IV (10 Hrs)	Project Selection and Appraisal: Brainstorming and concept evolution, Project selection and evaluation, Selection criteria and models, Types of appraisals, SWOT analysis, Cash flow analysis, Payback period, and Net present value.
UNIT-V (10 Hrs)	Project Implementation and Review: Identify various project types and apply appropriate management strategies for each. Forms of project organization – project planning – project control – human aspects of project management – prerequisites for successful project implementation – project review.
Textbooks:	
1.	Sharon Pande and Swapnalekha Basak, Human Resource Management, Text and Cases, Vikas Publishing, 2e, 2016.
2.	K. Nagarajan, Project Management, New Age International Publishers, 8e, 2017
Reference Books:	
1.	Subba Rao P: —Personnel and Human Resource Management-Text and Cases, Himalaya Publications, Mumbai, 2013.
2.	Prasanna Chandra, —Projects, Planning, Analysis, Selection, Financing Implementation and Review, Tata McGraw Hill Company Pvt. Ltd., New Delhi 1998
3.	Vasanth Desai, Project Management, 4th edition, Himalaya Publications 2018.
4.	Lalithabalakrishnan and Gowri, Project Management, Himalaya publishing house, New Delhi 2022
5.	K Aswathappa: —Human Resource and Personnel Management, Tata McGraw Hill, New Delhi, 2013
e-Resources	
1.	Robert L. Mathis, John H. Jackson, Manas Ranjan Tripathy, Human Resource Management, Cengage Learning 2016.
2.	Stewart R. Clegg, Torgeir Skyttermoen, Anne Live Vaagaasar, Project Management, Sage Publications, 1e, 2021.

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CS4102	PE	3	--	--	3	30	70	3 Hrs.
SOFTWARE ARCHITECTURE & DESIGN PATTERNS								
(Common to CSE, AIDS, CSBS, CSD)								
Pre-requisites: OOP,SE,OOAD								
Course Objectives: Students are expected								
1.	Understand the basic concepts to identify state behaviour of real world objects							
2.	Apply Object Oriented Analysis and Design concepts to solve complex problems							
3.	Construct various UML models using the appropriate notation for specific problem context							
4.	Design models to Show the importance of systems analysis and design in solving complex problems using case studies							
5.	Study Pattern Oriented approach for real world problems							
Course OutComes: At the end of the course students will be able to								
S. No	outcome							Knowledge Level
1.	Explain the concepts of design patterns and object-oriented development.							2
2.	Use object-oriented analysis techniques to gather requirements and identify conceptual classes and relationships in a system.							3
3.	Demonstrate structural design patterns such as Adapter, Bridge, Composite, Decorator, Facade, Flyweight, and Proxy.							3
4.	Illustrate the MVC architectural pattern in designing interactive systems.							3
5.	Apply distributed object system concepts including client-server architecture, Java RMI, and web services.							3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction: design pattern, Describing design patterns, the catalog of design pattern, organizing the catalog, how design patterns solve design problems, how to select a design pattern, how to use a design pattern. What is object oriented development-key concepts of object oriented design other related concepts, benefits and drawbacks of the paradigm							
UNIT-II (10 Hrs)	Analysis a System: Overview of the analysis phase, stage 1 gathering the requirements functional requirements specification, defining conceptual classes and relationships, using the knowledge of the domain Design and Implementation, discussions and further reading							
UNIT-III (10 Hrs)	Design Pattern Catalog: Structural patterns, Adapter, bridge, composite, decorator, facade, Flyweight, proxy.							
UNIT-IV	Interactive systems and the MVC architecture: Introduction The MVC architectural							

(10 Hrs)	pattern, analyzing a simple drawing program designing the system, designing of the subsystems, getting into implementation, implementing undo operation drawing incomplete items, adding a new feature pattern based solutions
UNIT-V (10 Hrs)	Designing with Distributed Objects: Client server system, java remote method invocation, implementing an object oriented system on the web, Web services (SOAP, Restful), Enterprise Service Bus
Text books:	
1.	Object Oriented Analysis, Design and Implementation, Brahma Dathan, Sarnath Rammath , Universities Press, 2013
2.	Design Patterns, Erich Gamma, Richard Helan, Ralph Johman, John Vlissides, PEARSON Publication, 2013
Reference books:	
1.	Frank Bachmann, Regine Meunier, Hans Rohnert “Pattern Oriented Software Architecture”, Volume 1, 1996.
2.	William J Brown et al., "Anti Patterns: Refactoring Software, Architectures and Projects in Crisis", John Wiley, 1998
E-Resources:	
1.	Object Oriented Analysis and Design By Prof. Partha Pratim Das, Prof. Samiran Chattopadhyay, Prof. Kausik Datta IIT Kharagpur https://onlinecourses.nptel.ac.in/noc21_cs57/preview
2.	Object Oriented System Development Using UML, Java And Patterns, IIT Kharagpur Prof. Rajib Mall https://nptel.ac.in/courses/106105224

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT4102	PE	3	--	--	3	30	70	3 Hrs.
DEEP LEARNING								
(For IT)								
Course Objectives: This course is designed to:								
1.	To learn fundamental concepts of deep learning, its evolution, and vision and learning principles.							
2.	To understand artificial neural networks, their training process, and techniques for improving performance.							
3.	To understand neural networks, CNNs, and RNNs with exposure to deep learning frameworks and implementation tools.							
4.	To understand applications of deep learning and key generative models used in modern AI systems.							
Course Outcomes: Upon the completion of the course students will be able to:								
S.No	Outcome							Knowledge Level
1.	Illustrate the basic concepts of deep learning, probabilistic models, biological vision and human learning.							K2
2.	Explain the architecture of artificial neural networks, activation and loss functions, training process, and performance improvement techniques.							K2
3.	Apply neural network concepts, deep learning frameworks, and convolutional neural network operations using programming tools for basic implementation.							K3
4.	Apply recurrent neural network concepts including LSTM, bidirectional RNNs, and encoder–decoder architectures for basic sequence modeling tasks.							K3
5.	Summarize the applications of deep learning and basic concepts of autoencoders and generative models.							K2
SYLLABUS								
UNIT-I	Fundamentals of Deep Learning: Artificial Intelligence, History of Machine learning: Probabilistic Modelling, Early Neural Networks, Kernel Methods, Random forests and Gradient Boosting machines. Introducing Deep Learning: Biological and machine vision, Human and machine learning.							
UNIT- II	Artificial Neural Networks (ANN): Architecture of ANN (Input, Hidden, Output Layers), Activation Functions: ReLU, Sigmoid, Tanh, Softmax. Loss Functions: MSE, Cross-Entropy, Hinge Loss. Backpropagation and Gradient Descent, Training Deep Networks, Improving Deep Networks: Dropout, Batch Normalization, Regularization (L1/L2).							

UNIT- III	Neural Networks: Anatomy of Neural Network, Introduction to Keras: Keras, Tensor Flow, Theano and CNTK, Setting up Deep Learning Workstation. Convolutional Neural Networks (CNN): Introduction to CNNs and Their Importance, Neural Network and Representation Learning, Convolutional Layers, Multichannel Convolution Operation, PyTorch Tensors: Deep Learning with PyTorch, CNN in PyTorch.
UNIT- IV	Recurrent Neural Networks (RNN): Introduction to RNN, Types of RNN, Limitations of RNN, Introduction to LSTM, LSTM architecture, Deep Recurrent Networks, Bidirectional RNNs, Encoder-Decoder Sequence-to-Sequence Architectures, Recursive Neural Networks.
UNIT- V	Interactive Applications of Deep Learning: Machine Vision, Natural Language processing, Generative Adversarial Networks(GAN), Deep Reinforcement Learning. Deep Learning Research: Autoencoders: Types of Autoencoders, Applications of Autoencoders, Deep Generative Models: Boltzmann Machines Restricted Boltzmann Machines, Deep Belief Networks.
Textbooks:	
1.	Deep Learning, Ian Goodfellow and Yoshua Bengio and Aaron Courville, MIT Press, 2016.
2.	Deep Learning with Python, Francois Chollet, December 2017, Manning Publications, ISBN: 9781617294433
3.	Deep Learning Illustrated: A Visual, Interactive Guide to Artificial Intelligence, Jon Krohn, Grant Beyleveld, Aglaé Bassens, September 2019, Addison-Wesley Professional, ISBN: 9780135116821
Reference Books:	
1.	Artificial Neural Networks, Yegnanarayana, B., PHI Learning Pvt. Ltd, 2009.
2.	Deep Learning with Python, François Chollet, Manning Publications, 2017.
3.	Neural Networks: A Systematic Introduction, Raúl Rojas, 1996.
4.	Deep Learning from Scratch, Seth Weidman, September 2019, O'Reilly Media, Inc., ISBN: 9781492041412
e-Resources	
1.	Swayam NPTEL: Deep Learning: https://onlinecourses.nptel.ac.in/noc22_cs22/preview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT4103	PE	3	--	--	3	30	70	3 Hrs.
COMPUTER VISION								
(Common to IT & CSD)								
Course Objectives: This course is designed to:								
1.	To understand the Fundamental Concepts related to sources, shadows and shading							
2.	To understand the Geometry of Multiple Views							
Course Outcomes: Upon the completion of the course students will be able to:								
S.No	Outcome							Knowledge Level
1.	Explain the fundamental principles of image formation and lighting							K2
2.	Demonstrate linear filtering techniques, frequency-domain concepts, edge detection methods, and texture representation models							K2
3.	Apply multi-view geometry concepts, stereoscopic reconstruction techniques, and clustering-based segmentation methods							K3
4.	Apply grouping and model fitting techniques, including Hough Transform and probabilistic methods							K3
5.	Implement geometric camera models, perform camera calibration using estimation techniques.							K3
SYLLABUS								
UNIT-I (10Hrs)	CAMERAS: Pinhole Cameras Radiometry – Measuring Light: Light in Space, Light Surfaces, Important Special Cases Sources, Shadows, And Shading: Qualitative Radiometry, Sources and Their Effects, Local Shading Models, Application: Photometric Stereo, Inter reflections: Global Shading Models Color: The Physics of Color, Human Color Perception, Representing Color, A Model for Image Color, Surface Color from Image Color.							
UNIT-II (8 Hrs)	Linear Filters: Linear Filters and Convolution, Shift Invariant Linear Systems, Spatial Frequency and Fourier Transforms, Sampling and Aliasing, Filters as Templates, Edge Detection: Noise, Estimating Derivatives, Detecting Edges Texture: Representing Texture, Analysis (and Synthesis) Using Oriented Pyramids, Application: Synthesis by Sampling Local Models, Shape from Texture.							

UNIT-III (10 Hrs)	The Geometry of Multiple Views: Two Views Stereopsis: Reconstruction, Human Stereopsis, Binocular Fusion, Using More Cameras Segmentation by Clustering: What Is Segmentation? Human Vision: Grouping and GetStalt, Applications: Shot Boundary Detection and Background Subtraction, Image Segmentation by Clustering Pixels, Segmentation by Graph-Theoretic Clustering.
UNIT-IV (10 Hrs)	Segmentation by Fitting a Model: The Hough Transform, Fitting Lines, Fitting Curves, Fitting as a Probabilistic Inference Problem, Robustness Segmentation and Fitting Using Probabilistic Methods: Missing Data Problems, Fitting, and Segmentation, The EM Algorithm in Practice, Tracking With Linear Dynamic Models: Tracking as an Abstract Inference Problem, Linear Dynamic Models, Kalman Filtering, Data Association, Applications and Examples
UNIT-V (8 Hrs)	Geometric Camera Models: Elements of Analytical Euclidean Geometry, Camera Parameters and the Perspective Projection, Affine Cameras and Affine Projection Equations Geometric Camera Calibration: Least-Squares Parameter Estimation, A Linear Approach to Camera Calibration, Taking Radial Distortion into Account, Analytical Photogrammetry, Case study: Mobile Robot Localization Model- Based Vision: Initial Assumptions, Obtaining Hypotheses by Pose Consistency, Obtaining Hypotheses by Pose Clustering, Obtaining Hypotheses Using Invariants, Verification, Case study: Registration In Medical Imaging Systems, Curved Surfaces and Alignment.
Textbooks:	
1.	David A. Forsyth and Jean Ponce: Computer Vision – A Modern Approach, PHI Learning (Indian Edition), 2009.
2.	Simon J.D Prince, Computer Vision: Models, Learning and Inference, 1st Edition, 2012.
Reference Books:	
1.	E. R. Davies: Computer and Machine Vision – Theory, Algorithms and Practicalities, Elsevier (Academic Press), 4th edition, 2013.
2.	R. C. Gonzalez and R. E. Woods “Digital Image Processing” Addison Wesley 2008. 3. Richard Szeliski “Computer Vision: Algorithms and Applications” Springer-Verlag London Limited 2011.
3.	Richard Szeliski “Computer Vision: Algorithms and Applications” Springer-Verlag London Limited 2011.
e-Resources	
1.	NPTEL LINK: https://onlinecourses.nptel.ac.in/noc22_ee48/preview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT4104	PE	3	--	--	3	30	70	3 Hrs.
BLOCKCHAIN TECHNOLOGY & APPLICATIONS								
(Common to IT,AIDS, CSBS)								
Course Objectives: This course is designed to:								
1.	Understand the fundamental concepts of blockchain technology, its architecture, and working principles.							
2.	Explain different types of blockchain systems and consensus mechanisms used in distributed environments.							
3.	Analyze public, private, and consortium blockchain platforms and their real-world implementations.							
4.	Explain and analyze cryptocurrency systems and smart contract mechanisms.							
5.	Analyze security, privacy challenges, and performance issues in blockchain technology.							
Course Outcomes: Upon the completion of the course students will be able to:								
S.No	Outcome							Knowledge Level
1.	Explain blockchain fundamentals, architecture, and cryptographic concepts							K2
2.	Explain smart contracts and cryptocurrency transaction mechanisms in blockchain platforms							K3
3.	Analyze private and consortium blockchain systems and permissioned blockchain algorithms							K2
4.	Analyze security, privacy challenges and applications of blockchain in emerging real-world scenarios.							K3
5.	Analyze blockchain case studies and implement blockchain applications using python and hyperledger fabric.							K3
SYLLABUS								
UNIT-I	Fundamentals of Blockchain: Introduction, Origin of Blockchain, Blockchain Solutions, Components of Blockchain, Block in a Blockchain, The Technology and the Future. Blockchain Types and Consensus Mechanism: Introduction, Decentralization and Distribution, Types of Blockchain, Consensus Protocol. Cryptocurrency: Bitcoin, Altcoin and Token: Introduction, Bitcoin and the Cryptocurrency, Cryptocurrency Basics, Types of Cryptocurrencies, Cryptocurrency Usage.							
UNIT-II	Public Blockchain System: Introduction, Public Blockchain, Popular Public Blockchains, The Bitcoin Blockchain, Ethereum Blockchain. Smart Contracts: Introduction, Smart Contract, Characteristics of a Smart Contract, Types of Smart Contracts, Types of Oracles, Smart Contracts in Ethereum, Smart Contracts in Industry.							

UNIT-III	Private Blockchain System: Introduction, Key Characteristics of Private Blockchain, Private Blockchain, Private Blockchain Examples, Private Blockchain and Open Source, E-commerce Site Example, Various Commands (Instructions) in E-commerce Blockchain, Smart Contract in Private Environment, State Machine, Different Algorithms of Permissioned Blockchain, Byzantine Fault, Multichain. Consortium Blockchain: Introduction, Key Characteristics of Consortium Blockchain, Need of Consortium Blockchain, Hyperledger Platform, Overview of Ripple, Overview of Corda.
UNIT-IV	Security in Blockchain: Introduction, Security Aspects in Bitcoin, Security and Privacy Challenges of Blockchain in General, Performance and Scalability, Identity Management and Authentication, Regulatory Compliance and Assurance, Safeguarding Blockchain Smart Contract (DApp), Security Aspects in Hyperledger Fabric. Applications of Blockchain: Introduction, Blockchain in Banking and Finance, Blockchain in Education, Blockchain in Energy, Blockchain in Healthcare, Blockchain in Real-estate, Blockchain in Supply Chain, The Blockchain and IoT. Limitations and Challenges of Blockchain.
UNIT-V	Blockchain Case Studies: Case Study 1 – Retail, Case Study 2 – Banking and Financial Services, Case Study 3 – Healthcare, Case Study 4 – Energy and Utilities. Blockchain Platform using Python: Introduction, Learn How to Use Python Online Editor, Basic Programming Using Python, Python Packages for Blockchain. Blockchain platform using Hyperledger Fabric: Introduction, Components of Hyperledger Fabric Network, Chain codes from Developer.ibm.com, Blockchain Application Using Fabric Java SDK.
Textbooks:	
1.	“Blockchain Technology”, Chandramouli Subramanian, Asha A.George, Abhilasj K A, Meena Karthikeyan , Universities Press.
2.	Blockchain for Business, Jai Singh Arun, Jerry Cuomo, Nitin Gaur, Pearson Addition Wesley
Reference Books:	
1.	Blockchain Blue print for Economy, Melanie Swan, SPD Oreilly.
e-Resources:	
1	NPTEL: Blockchain Architecture Design and Use Cases – https://nptel.ac.in
2	Ethereum Documentation – https://ethereum.org
3	Hyperledger Documentation – https://hyperledger.org

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CS4107	PE	3	--	--	3	30	70	3 Hrs.
AGILE METHODOLOGIES								
(Common to CSE,AI ML AIDS, CSBS, IT)								
Pre-requisites: Software Engineering								
Course Objectives: Students are expected								
1.	Gain deep comprehension of Agile philosophy, principles, and values to support high-quality software delivery.							
2.	Employ different Agile frameworks (Scrum, XP, Lean, Kanban) to real-world projects for improved adaptability, productivity, and customer satisfaction.							
Course Out Comes: At the end of the course students will be able to								
S. No	OUTCOME							KL
1.	Apply agile values and principles to foster team collaboration and adaptability in projects.							K3
2.	Demonstrate the 12 agile principles to direct project delivery in changing contexts.							K3
3.	Use Scrum roles, events, and artifacts to oversee iterative development cycles.							K3
4.	Model XP practices to boost software quality and team responsiveness.							K3
5.	Determine Lean and Kanban principles to detect waste and streamline development workflows.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Learning Agile: Agile, Getting Agile into your brain, Understanding Agile values: No Silver Bullet, Agile to the Rescue. A fractured perspective, The Agile Manifesto, Understanding the Elephant, Where to Start with a New Methodology.							
UNIT-II (10 Hrs)	The Agile Principles: The 12 Principles of Agile Software, The Customer Is Always Right, Delivering the Project, Better Project Delivery for the Ebook Reader Project. Communicating and Working Together, Project Execution—Moving the Project Along, Constantly Improving the Project and the Team. The Agile Project: Bringing All the Principles Together							
UNIT-III (10 Hrs)	SCRUM and Self-Organizing Teams: The Rules of Scrum, Act I: I Can Haz Scrum, Everyone on a Scrum Team owns the Project, Status Updates Are for Social Networks!, The Whole Team Uses the Daily Scrum, Feedback and the Visibility-Inspection-Adaptation Cycle, The Last Responsible Moment, Sprinting into a Wall, Sprints, Planning, and Retrospectives. Scrum Planning And Collective Commitment: Not Quite Expecting the Unexpected, User							

	Stories, Velocity, and Generally Accepted Scrum Practices, Victory Lap, Scrum Values Revisited.
UNIT-IV (10 Hrs)	XP And Embracing Change: Going into Overtime, The Primary Practices of XP, The Game Plan Changed, but We're Still Losing, The XP Values Help the Team Change Their Mindset, An Effective Mindset Starts with the XP Values, The Momentum Shifts, Understanding the XP Principles Helps You Embrace Change. XP, Simplicity, and Incremental Design: Code and Design, Make Code and Design Decisions at the Last Responsible Moment, Final Score.
UNIT-V (10 Hrs)	Lean, Eliminating Waste, and Seeing the whole: Lean Thinking, Creating Heroes and Magical Thinking. Eliminate Waste, Gain a Deeper Understanding of the Product, Deliver As Fast As Possible. Kanban, Flow, and Constantly Improving: The Principles of Kanban, Improving Your Process with Kanban, Measure and Manage Flow, Emergent Behavior with Kanban. The Agile Coach: Coaches Understand Why People Don't Always Want to Change. The Principles of Coaching.
Textbook:	
1.	Andrew Stellman, Jennifer Greene, Learning Agile, O'Reilly, 2015.
Reference books:	
1.	Andrew Stellman, Jennifer Green, Head first Agile, O'Reilly, 2017.
2.	Rubin K , Essential Scrum : A Practical Guide To The Most Popular Agile Process, Addison-Wesley, 2013
E-Resources:	
1.	https://www.agilealliance.org
2.	https://www.scrum.org
3.	https://www.scrumtrainingseries.com
4.	https://www.agilemanifesto.org
5.	https://www.scrumguides.org

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT4106	PE	3	--	--	3	30	70	3 Hrs.
BIG DATA ANALYTICS								
(For IT)								
Course Objectives: This course is designed to:								
1.	Provide an overview of an exciting growing field of big data analytics.							
2.	Introduce the tools required to manage and analyze big data like Hadoop, NoSQL, Map Reduce, HIVE, Cassandra, Spark							
3.	Teach the fundamental techniques and principles in achieving big data analytics with scalability and streaming capability.							
4.	Optimize business decisions and create competitive advantage with Big Data analytics							
Course Outcomes: Upon the completion of the course students will be able to:								
S.No	Outcome							Knowledge Level
1.	Understand the existing technologies and the need of distributed files Systems to analyze the Big Data							K2
2.	Develop the ability to design, analyze, and implement scalable NoSQL database systems using various data models, distribution strategies, and tools like Cassandra.							K3
3.	Explore the features of HDFS and Map Reduce to handle the Big Data and identify the need of interfaces to perform I/O operations in Hadoop							K3
4.	Provide skills to use Hive and Apache Spark for efficient big data storage, processing, and analysis							K3
5.	Enable skills to optimize Spark performance and build robust real- time data processing applications							K3
SYLLABUS								
UNIT-I	Big data, convergence of key trends, unstructured data, industry examples of big data, web analytics, big data and marketing, fraud and big data, risk and big data, credit risk management, big data and algorithmic trading, big data and healthcare, big data in medicine, advertising and big data, big data technologies, introduction to Hadoop, open source technologies, cloud and big data.							
UNIT-II	Introduction to NoSQL, aggregate data models, aggregates, key-value and document data models, relationships, graph databases, schema less databases, materialized views, distribution models, sharding, master-slave replication, peer- peer replication, sharding and replication, working with Cassandra, Table creation, loading and reading data							

UNIT-III	HDFS Design: HDFS Architecture, Building Blocks, HDFS features: Replication, Rack Awareness, fault tolerance with data replication, High Availability, Data Locality, HDFS Read & Write. Analyzing the Data with Hadoop: Hadoop Streaming, Java Interfaces to Map Reduce, Anatomy of a Map Reduce Job Run (classic Map Reduce and Yarn) Basic programs of Hadoop Map Reduce: Map Reduce word count example, Matrix multiplication using Map Reduce.
UNIT-IV	Hive: Introduction to Hive, Hive Architecture, Hive Data Types and file formats, HiveQL data definition, HiveQL data manipulation. Apache spark- Advantages over Hadoop, lazy evaluation, in memory processing, DAG, Spark context, Spark Session, RDD, Transformations- Narrow and Wide, Actions, Data frames, RDD to Data frames, Catalyst optimizer, Data Frame Transformations, working with Dates and Timestamps, working with Nulls in Data, working with Complex Types, working with JSON, Grouping, Window Functions, Joins, Data Sources, Broadcast Variables, Accumulators. Spark on YARN
UNIT-V	Spark - Performance Tuning, Stream Processing Fundamentals, Event-Time and State Full Processing - Event Time, State full Processing, Windows on Event Time- Tumbling Windows, Handling Late Data with Watermarks, Dropping Duplicates in a Stream, Structured Streaming Basics - Core Concepts, Structured Streaming in Action, Transformations on Streams, Input and Output.
Textbooks:	
1.	Big Data, Big Analytics: Emerging, Michael Minnelli, Michelle Chambers, Ambiga Dhiraj, 1 st edition ,2013
2.	SPARK: The Definitive Guide, Bill Chambers & Matei Zaharia, O'Reilley, 2018- first Edition.
3	Business Intelligence and Analytic Trends for Today's Businesses", Wiley, First edition-2013.
4	P. J. Sadalage and M. Fowler, "NoSQL Distilled: A Brief Guide to the Emerging World Polyglot Persistence", Addison-Wesley Professional, 2012
5	Tom White, "Hadoop: The Definitive Guide", Third Edition, O'Reilley, 2012
Reference Books:	
1.	"Hadoop Operations", O'Reilley, Eric Sammer, First Edition -2012.
2.	"Programming Hive", O'Reilley, E. Capriolo, D. Wampler, and J. Rutherglen, 2012.
3.	"HBase: The Definitive Guide", O'Reilley, Lars George, September 2011: First Edition.
4.	"Cassandra: The Definitive Guide", O'Reilley, Eben Hewitt, 2010.
5.	"Programming Pig", O'Reilley, Alan Gates, October 2011: First Edition

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT4107	PE	3	--	--	3	30	70	3 Hrs.
MOBILE COMPUTING								
(For IT)								
Course Objectives: This course is designed to:								
1.	Introduce the basic concepts and principles in mobile computing, explore both theoretical and practical issues of mobile computing and understand latest network architecture and its interfaces							
2.	Provide an opportunity for the students to understand the key components and technologies involved and to gain hands-on experience in building mobile applications.							
Course Outcomes: Upon the completion of the course students will be able to:								
S.No	Outcome							Knowledge Level
1.	Explain the fundamental concepts of mobile communication, including guided and unguided transmission, signal propagation, antennas, modulation techniques, mobile computing architecture, and the characteristics and limitations of mobile devices and systems.							K2
2.	Describe the architecture and operation of GSM, GPRS, and wireless communication technologies (2G to 6G), including multiplexing, spread spectrum, and medium access control techniques.							K2
3.	Apply Mobile IP, TCP/IP, and MANET concepts to demonstrate packet delivery, handover management, routing, and basic security in mobile and ad-hoc networks.							K3
4.	Explain synchronization mechanisms in mobile computing systems, including usage models, data synchronization rules, conflict resolution strategies, mobile agent design (aglets), and application server concepts.							K2
5.	Apply the principles of wireless LANs, IEEE 802.11 protocol layers, and Wireless Application Protocol (WAP) to demonstrate the functioning of mobile internet communication systems.							K3
SYLLABUS								

UNIT- I	Mobile Communications: An Overview- Mobile Communication-guided transmission, unguided transmission- signal propagation frequencies, antennae, modulation, modulation methods and standards for voice-oriented data communication standards, modulation methods and standards for data and voice communication, mobile computing- novel applications and limitations, mobile computing architecture, mobile system networks. Mobile devices and systems: Cellular networks and frequency reuse, Mobile smart phones, Smart mobiles and systems, handheld pocket computers, Handheld devices, Smart systems, Limitations of mobile devices
UNIT-II	GSM and other 2G Architectures: GSM-services and system architecture, Radio interfaces of GSM, Protocols of GSM, Localization, Call handling, GPRS system architecture. Wireless medium access control, CDMA, 3G, 4G , 5G and 6G. Communication: Modulation, Multiplexing, Controlling the medium access, Spread spectrum, Coding methods, IMT-2000/3G wireless communication standards, WCDMA 3G communication standards, CDMA 3G communication standards, Broadband wireless access, 4G networks.
UNIT-III	Mobile IP Network layer: IP and Mobile IP network layers: OSI layer functions, TCP/IP and Internet protocol, Mobile internet protocol; Packet delivery and Handover Management; Location Management: Agent Discovery; Mobile TCP Introduction to Mobile Adhoc network: fixed infrastructure architecture, MANET infrastructure architecture; MANET: properties, spectrum, applications; Security in Ad-hoc network; Wireless sensor networks; sensor network applications.
UNIT-IV	Synchronization: Synchronization in mobile computing systems, Usage models for Synchronization in mobile application, Domain-dependant specific rules for data synchronization, Personal information manager, synchronization and conflict resolution strategies, synchronizer; Mobile agent: mobile agent design, aglets; Application Server
UNIT-V	Mobile Wireless Short Range Networks and Mobile Internet: Wireless networking and wireless LAN, Wireless LAN (WLAN) architecture, IEEE 802.11 protocol layers, Wireless application protocol (WAP)-WAP1.1 architecture, wireless datagram protocol (WDP), Wireless Transport Layer Security (WTLS), wireless transaction and session layers, wireless application environment.
Textbooks:	
1	Raj Kamal, "Mobile Computing," second edition, Oxford.
2	UWE Hansmann, Lothar Merk, Martin S. Nocklous, Thomas Stober, "Principles of Mobile Computing," Second Edition, Springer
Reference Books:	

1	Jochen Schiller, “Mobile Communications”, Addison-Wesley, Second Edition, 2009
2	Asoke K Talukder, Hasanahmed, Roopa R Yavagal, “Mobile Computing, Technology Applications and Service Creation” Second Edition, Mc Graw Hill.



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT4108	PE	3	--	--	3	30	70	3 Hrs.
CYBER PHYSICAL SYSTEMS								
(Common to IT, CSE, CSD)								
Course Objectives: This course is designed to:								
1.	To introduce principles of cyber-physical systems, including embedded sensing, actuation, and control.							
2.	To teach formal methods for modeling dynamic, real-time systems, such as state- space models and hybrid automata							
3.	To enable students to apply AI techniques, control theory, and data analytics in practical engineering problems.							
Course Outcomes: Upon the completion of the course students will be able to:								
S.No	Outcome						Knowledge Level	
1.	Understand and apply symbolic synthesis techniques to model, analyze, and design controllers for Cyber-Physical Systems						K3	
2.	Analyze security requirements, attack models, and countermeasures in Cyber-Physical Systems,						K3	
3.	Understand and address synchronization challenges in distributed Cyber-Physical Systems						K2	
4.	Apply real-time scheduling techniques for Cyber Physical Systems						K3	
5.	Understand and apply model integration techniques in Cyber-Physical Systems,						K3	
SYLLABUS								
UNIT-I	Symbolic Synthesis for Cyber-Physical Systems: Introduction and Motivation, Basic Techniques - Preliminaries, Problem Definition, Solving the Synthesis Problem, Construction of Symbolic Models, Advanced Techniques: Construction of Symbolic Models, Continuous Time Controllers, Software Tools							
UNIT-II	Security of Cyber-Physical Systems: Introduction and Motivation, Basic Techniques - Cyber Security Requirements, Attack Model, Countermeasures, Advanced Techniques - System Theoretic Approaches							
UNIT-III	Synchronization in Distributed Cyber-Physical Systems: Challenges in Cyber-Physical Systems, A Complexity-Reducing Technique for Synchronization, Formal Software Engineering, Distributed Consensus Algorithms, Synchronous Lockstep Executions, TimeTriggered Architecture, Related Technology, Advanced Techniques							

UNIT-IV	Real-Time Scheduling for Cyber-Physical Systems: Introduction and Motivation, Basic Techniques – Scheduling with Fixed Timing Parameters, Memory Effects, Multiprocessor/Multicore Scheduling, Accommodating Variability and Uncertainty
UNIT-V	Model Integration in Cyber-Physical Systems: Introduction and Motivation, Causality, Semantic Domains for Time, Interaction Models for Computational Processes, Semantics of CPS DSMLs, Advanced Techniques, ForSpec, The Syntax of CyPhyML, Formalization of Semantics, Formalization of Language Integration.
Textbooks:	
1.	Raj Rajkumar, Dionisio De Niz, and Mark Klein, Cyber-Physical Systems, AddisonWesley Professional, 2016
2.	Rajeev Alur, Principles of Cyber-Physical Systems, MIT Press
Reference Books:	
1.	E.A.Lee, Sanjit Seshia, Introduction to Embedded Systems: A Cyber-Physical Systems Approach, MIT Press
2.	Andre Platzer, Logical Foundations of Cyber-Physical Systems, (2e), Springer Publishing, 2018



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT4110	SEC	--	1	2	2	30	100	3 Hrs
PROMPT ENGINEERING								
(for IT)								
Course Objectives: This course is designed to:								
1.	Apply iterative prompting for clarity and context.							
2.	Create varied prompts to steer model outputs.							
3.	Construct chain-of-thought and structured prompts.							
4.	Develop retrieval-augmented pipelines to ground outputs.							
5.	Evaluate LLM agents and multimodal apps for ethics and robustness.							
Course Outcomes: Upon the completion of the course students will be able to:								
S.No	Outcome							Knowledge Level
1.	Explain core principles of prompt engineering, its distinction from fine-tuning, and the iterative prompting lifecycle.							K2
2.	Apply advanced prompt patterns, such as few-shot, role-based, negative prompting, and constraint specification, to generate consistent and targeted LLM outputs.							K3
3.	Construct structured prompts for formatted outputs (e.g., JSON, tables) and chain-of-thought reasoning to solve complex tasks through decomposition.							K3
4.	Develop basic retrieval-augmented generation (RAG) pipelines using LangChain, including indexing, embeddings, and LCEL chains, to ground LLM responses in external data.							K2
5.	Evaluate LLM agents, multimodal applications, and prompts for ethics, robustness, security risks (e.g., prompt injection), and performance using manual and automated techniques.							K3
SYLLABUS								
UNIT-I (10Hrs)	Foundations of Prompt Engineering: Definition of prompt engineering, Distinction between prompt engineering and model fine-tuning, Motivation and benefits of prompt							

	engineering, Core principles of effective prompt design, Anatomy of a prompt, Setting up the Python environment for LLM interaction, Iterative prompting lifecycle, Common prompt pitfalls and remediation. Lab Experiments: 1. Environment & Connectivity: Install required packages (e.g., transformers, openai); securely configure the API key; run a simple “Hello, world” prompt to verify model access. 2. Baseline vs. Enhanced Prompts: Execute a naïve prompt (“Write a one-paragraph bio of Ada Lovelace.”) and an enhanced prompt that adds role framing, specificity, and explicit format instructions; compare both outputs for relevance, completeness, and style. 3. Iterative Refinement on a Simple Task: Summarize the plot of the Shakespearean play Romeo and Juliet in two sentences through three rounds of prompt tweaking: a. Minimal instruction. b. Addition of length and style constraints c. Specification of key content elements (setting and theme) Document how each iteration changes and improves the result. 4. Diagnosing Prompt Failures & Edge Cases: Craft a vague or contradictory prompt; analyze the failure mode (ambiguity, missing context, or format errors); refine the prompt by adding examples or clarifying instructions.
UNIT-II (8 Hrs)	Advanced Prompt Patterns & Techniques: Enhanced prompt anatomy: contextual detail and explicit output specifications, Few-shot in-context prompting, Prompt structuring and template design, Role-based prompting to establish personas or system behavior, Negative prompting to filter or suppress undesired content, Constraint specification and instruction enforcement (e.g., length, format), Iterative prompt refinement and optimization. Lab Experiments: 1. Few-Shot vs. Zero-Shot Comparison: Design and execute a zero-shot prompt and a few-shot prompt (with 2–3 exemplar input-output pairs) for a chosen text task (e.g., sentiment classification or translation); compare outputs for accuracy, consistency, and adherence to examples. 2. Role-Based & Negative Prompting: Craft a role-based prompt to establish a specific persona (e.g., “You are a financial advisor...”); then create a negative prompt to suppress undesired content (e.g., “Do not mention any brand names”); evaluate how each influences the model’s response. 3. Constraint Specification & Iterative Refinement: Select an open-ended task (e.g., summarizing a technical article); issue a basic prompt; identify failures in length or format; refine the prompt by adding explicit constraints (word count, bullet format, etc.); document improvements over two refinement cycles.

UNIT-III (10 Hrs)	Structured Output & Reasoning Techniques: Importance of structured outputs for real-world applications, Prompting for specific formats (lists, tables, Markdown), Generating valid JSON and YAML via explicit instructions, Eliciting chain-of-thought reasoning in zero-shot prompts, Decomposing complex tasks into manageable sub-tasks Lab Experiments: 1. Structured Format Prompting: Instruct the model to output information as bullet lists and Markdown tables (e.g., “List three benefits of daily exercise in a Markdown table with columns ‘Benefit’ and ‘Description.’”); verify the output matches the requested structure. 2. JSON/YAML Generation: Provide a brief dataset description (e.g., three books with title, author, publication year) and prompt the model to produce valid JSON or YAML; use a parser to validate syntax and refine the prompt if errors occur. 3. Chain-of-Thought & Task Decomposition: Present a multi-step problem (e.g., a logic puzzle) and apply zero-shot CoT prompting (e.g., “Let’s think step by step. Explain your reasoning before the final answer.”); separately, decompose the problem into sequential sub-questions, collect partial answers, combine them, and compare accuracy against a direct-answer baseline.
UNIT-IV (10 Hrs)	Retrieval-Augmented Generation & LangChain Workflows: Limitations of LLM internal knowledge, Need for external data sources, Introduction to Retrieval-Augmented Generation (RAG), Overview of RAG architecture (indexing vs. retrieval + generation), Getting started with LangChain for LLM applications, Basics of LangChain Expression Language (LCEL), Simplified indexing pipeline: document loading & text splitting, Fundamentals of embeddings and vector stores, Building a basic retrieval- generation pipeline with an LCEL chain Lab Experiments: 1. Building a Simple LCEL Chain: Create a minimal LCEL script that accepts a fixed instruction (e.g., “Summarize this text: ...”), passes it to an LLM, and prints the result; verify end-to-end execution. 2. Basic Data Indexing for RAG: Load a small collection of documents; split into uniform chunks (e.g., 200 tokens); generate embeddings for each chunk; store them in an in-memory vector store; inspect for consistency. 3. Constructing & Running a Basic RAG Chain: Build a pipeline that: a. Receives a user query b. Retrieves the top-k relevant chunks c. Constructs a combined prompt with context + query d. Send it to the LLM e. Returns the answer e. Test with sample queries and compare factual accuracy against a prompt without retrieval.

UNIT-V (8 Hrs)	Agents, Multimodal AI & Ethical Evaluation: Introduction to LLM agents and their basic architecture, Overview of multimodal AI models (VLMs), Prompting for text-to-image generation and image understanding, Importance of prompt evaluation beyond subjective judgment, Manual evaluation techniques (heuristic checks for accuracy, relevance, format), Introduction to “LLM-as-Judge” for automated evaluation, Security considerations (prompt injection, sensitive-information risks), Prompt-based mitigation strategies for safety and robustness, Ethical concerns (bias, misinformation, data privacy), Brief exploration of UI frameworks (Streamlit/Gradio) for deploying prompt-driven apps, Adapting to the evolving nature of prompt engineering through continuous learning Lab Experiments: 1. Building a Simple LLM Agent: Register a tool (e.g., a calculator function) and craft prompts that instruct the agent to invoke it when required; implement using LangChain or a function-calling API; test on queries requiring tool execution. 2. Multimodal Prompting Exploration: Generate images from detailed text prompts; feed one generated image into an image-understanding model or API with an appropriate prompt; compare the returned caption to the original prompt to evaluate alignment. 3. Prompt Evaluation & Ethics Workshop: a. Select two existing prompts and generate multiple outputs; apply manual heuristic checks for accuracy, relevance, and format compliance. b. Use an “LLM-as-Judge” prompt (e.g., “Rate these outputs on a scale of 1–5 for clarity and correctness.”) to automate evaluation. c. Design a prompt- injection test (e.g., “Ignore previous instructions...”), observe the response, then refine system prompts to mitigate the vulnerability.
Textbooks:	
1.	James Phoenix, Mike Taylor, “Prompt Engineering for Generative AI”, O’Reilly, To Release in May 2024 https://www.oreilly.com/library/view/prompt-engineering-for/9781098153427/
2.	Gilbert Mizrahi, “Unlocking the Secrets of Prompt Engineering: Master the Art of Creative Language Generation to Accelerate Your Journey from Novice to Pro”, January 2024 https://www.packtpub.com/en-in/product/unlocking-the-secrets-of-prompt-engineering9781835083833
Reference Books:	
1.	Michael Ferguson, “Prompt Engineering: The Future of Language Generation”, January 2023 https://books.apple.com/us/book/prompt-engineering-the-future-of-languagegeneration/id6445529200
2.	“Prompt Engineering Guide”, https://www.promptingguide.ai/
3.	“Prompt Engineering for Generative AI”, Google, https://developers.google.com/machinelearning/resources/prompt-eng

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MC4101	MC	2	--	--	--	30	-	3 Hrs.
CONSTITUTION OF INDIA								
(Common to all)								
Course Objectives: This course is designed to:								
1	To provide understanding of the fundamental principles, sources, and key features of the Indian Constitution.							
2	To familiarize students with the structure and functioning of Union, State, and Local Governments in India.							
3	To develop awareness about the roles of constitutional bodies like the Election Commission and welfare commissions in promoting democracy and social justice.							
Course Outcomes: Upon the completion of the course students will be able to:								
S.No	Outcome							Knowledge Level
1.	Apply the concepts of the Indian Constitution to interpret its features such as Citizenship, Preamble, Fundamental Rights and Duties, and Directive Principles of State Policy.							K3
2.	Apply the structure and functions of the Union Government to analyze the roles of executive, legislature, and judiciary in the Indian administrative system.							K3
3.	Apply the structure and functions of the State Government to interpret the roles of the Governor, Chief Minister, and State Secretariat in administration.							K3
4.	Apply the structure and functions of local administration to interpret the roles of district, municipal, and Panchayati Raj institutions in grass roots governance.							K3
5.	Apply the roles and functions of the Election Commission and various welfare commissions to interpret their significance in ensuring democratic governance and social justice.							K3
SYLLABUS								
UNIT-I (10Hrs)	Introduction to Indian Constitution: Constitution meaning of the term, Indian Constitution - Sources and constitutional history, Features - Citizenship, Preamble, Fundamental Rights and Duties, Directive Principles of State Policy.							
UNIT-II (8 Hrs)	Union Government and its Administration Structure of the Indian Union: Federalism, Centre-State relationship, President: Role, power and position, PM and Council of ministers, Cabinet and Central Secretariat, Lok Sabha, Rajya Sabha, The Supreme Court and High Court: Powers and Functions.							
UNIT-III (10 Hrs)	State Government and its Administration Governor - Role and Position - CM and Council of ministers, State Secretariat: Organisation, Structure and Functions.							

UNIT-IV (10 Hrs)	Local Administration - District's Administration Head - Role and Importance, Municipalities — Mayor and role of Elected Representative - CEO of Municipal Corporation Pachayati Raj: Functions PRI: Zilla Panchayat, Elected officials and their roles, CEO Zilla Panchayat: Block level Organizational Hierarchy - (Different departments), Village level - Role of Elected and Appointed officials - Importance of grass root democracy.
UNIT-V (8 Hrs)	Election Commission: Election Commission- Role of Chief Election Commissioner and Election Commissionerate State Election Commission: Functions of Commissions for the welfare of SC/ST/OBC and women.
Textbooks:	
1.	Durga Das Basu, Introduction to the Constitution of India, Prentice — Hall of India Pvt. Ltd..New Delhi.
2.	Subash Kashyap, Indian Constitution, National Book Trust.
Reference Books:	
1.	J.A. Siwach, Dynamics of Indian Government & Politics, Sterling Publishers.
2.	D.C. Gupta, Indian Government and Politics, S Chand.
3.	H.M.Sreevai, Constitutional Law of India, 4th edition in 3 volumes, Universal Law Publication.
4.	M.V. Pylee, Indian Constitution Durga Das Basu, Human Rights in Constitutional Law, Prentice Hall of India Pvt. Ltd.. New Delhi.
5.	Noorani, A.G., (South Asia Human Rights Documentation Centre), Challenges to Civil Right), Challenges to Civil Rights Guarantees in India, Oxford University Press 2012.
e-Resources	
1.	https://onlinecourses.swayam2.ac.in/e-learning/preview/imb26_mg151
2.	https://onlinecourses.nptel.ac.in/noc24_lw05/preview
3.	https://nptel.ac.in/courses/129106750



**SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE
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Regulation: R23			IV / IV - B.Tech. II - Semester						
INFORMATION TECHNOLOGY									
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)									
Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23IT4201	Project and Internship	PR	-	-	24	12	60	140	200
TOTAL			0	0	24	12	60	140	200



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Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT4201	PR	-	-	24	12	60	140	3 Hrs.
PROJECT WORK								
(For IT)								
Course Objectives: This course is designed to:								
1	To enable students to independently research, analyze, and formulate a well-defined engineering problem by reviewing contemporary literature and technical databases.							
2	To guide students in conceptualizing, designing, and optimizing innovative solutions, components, or processes while balancing realistic socio-economic and technical constraints.							
3	To train students to build, test, and troubleshoot physical prototypes or software systems through the self-taught application of modern engineering tools and simulation software.							
4.	To encourage students to evaluate the environmental, safety, legal, and cultural impacts of their engineering solutions while maintaining professional ethics.							
5.	To nurture collaborative teamwork, project resource management, technical report writing, and effective multi-media presentation skills.							
Course Outcomes: Upon the completion of the course students will be able to:								
S.No	Outcome							Knowledge Level
1.	Identify and formulate a complex engineering problem by reviewing literature and conducting feasibility studies.							K4
2.	Design and develop an innovative, safe, and sustainable solution, component, or process using modern engineering tools.							K5
3.	Implement and test the designed solution, analyzing the resulting data to draw valid engineering conclusions.							K4
4.	Evaluate the impact of the project solution on society, environment, and health, incorporating ethical considerations.							K5
5.	Communicate effectively through technical reports, presentations, and project demonstrations.							K3
6	Demonstrate teamwork, leadership, and project management skills, adhering to ethical practices and financial constraints.							K3
*The objective of the Project Work is to enable the student to take up investigative study in the broad field of, Information Technology either fully theoretical/practical or involving both theoretical and practical work to be assigned by the Department on an individual basis or a group of students, under the guidance of a Supervisor. This is expected to provide a good initiation for the student(s) in R&D work.								
The assignment to normally include:								
a) Survey and study of published literature on the assigned topic.								
b) Working out a preliminary approach to the problem relating to the assigned topic.								
c) Conducting preliminary Analysis/Modeling/Simulation/Experiment/Design/ Feasibility.								
d) Preparing a written report on the study conducted for presentation to the department.								
e) Final Seminar, as oral Presentation before a departmental committee.								